

Time Beyond Light: What If Something Could Travel Faster Than Light?

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Date: January 2026

Keywords: Time, Relativity, Faster-than-Light, Spacetime, Causality, Entropy, Emergent Time

JEL Classification: D80, Z19

Abstract

Time is strange. In Newton's world, it's absolute, universal, ticking the same for everyone. In Einstein's relativity, it bends and stretches depending on speed and gravity. But what if something could go faster than light? What happens to time then?

This paper is a conceptual exploration purely thought experiment of what time could mean if an object moves faster than light. We examine spacetime intervals, proper time, and causality, and explore alternative ways time could exist emergent from correlations, information, or entropy rather than just motion.

The goal isn't to claim faster-than-light travel is possible; it's to question what time really is when the rules of relativity reach their limits

1. Introduction

Time is one of the most fundamental yet conceptually elusive elements of physical theory. In classical Newtonian mechanics, time is treated as absolute and universal, progressing uniformly for all observers regardless of motion or location. With the advent of Einstein's theory of relativity, this notion was fundamentally revised: time became relative, dependent on an observer's velocity and gravitational environment. In quantum mechanics, time occupies yet another role, appearing as an external parameter rather than a dynamical observable, governing the evolution of quantum states without itself being subject to quantization.

Despite these differing treatments, modern physical theories share a common underlying assumption: the speed of light in vacuum represents an ultimate and insurmountable limit for the transmission of matter, energy, and information. The structure of spacetime, the definition of proper time, and the preservation of causality are all built upon this constraint.

This paper adopts a purely conceptual and exploratory perspective to examine a provocative question: if one were to hypothetically consider motion faster than the speed of

light, how would the concept of time be understood? Rather than asserting the physical feasibility of faster-than-light travel, the aim is to investigate the implications such a scenario would have for established notions of time, spacetime intervals, and causal ordering.

By treating faster-than-light motion as a thought experiment, this work seeks to illuminate the boundaries of relativistic time and to explore whether alternative interpretations—such as emergent, relational, or entropy-based notions of time—might provide meaningful ways to describe temporal ordering when conventional definitions break down.

2. A Quick Recap of Time in Physics

2.1 Newtonian Time

In Newtonian mechanics, time is absolute and independent of physical processes. The equations of motion:

$$F = ma, \quad v = dx/dt, \quad a = dv/dt$$

treat time as a universal parameter that progresses identically for all observers.

2.2 Relativistic Time

Special relativity introduces velocity-dependent time dilation:

$$\Delta t = \Delta \tau / \sqrt{1 - v^2/c^2}$$

As velocity approaches the speed of light, time slows relative to a stationary observer. At $v = c$, the expression becomes undefined, indicating a breakdown of the theory at or beyond this limit.

2.3 Time in Quantum Mechanics

In quantum mechanics, time is not treated as an operator but as an external parameter:

$$i\hbar \partial \Psi / \partial t = \hat{H} \Psi$$

This creates a conceptual mismatch with relativistic treatments of time.

3. The Thought Experiment: Faster-Than-Light Travel

3.1 Spacetime Interval

The Minkowski spacetime interval is given by:

$$ds^2 = -c^2 dt^2 + dx^2$$

For FTL motion, intervals become spacelike ($ds^2 > 0$), leading to imaginary proper time:

$$\Delta\tau = \sqrt{-ds^2}/c$$

This implies that conventional notions of time do not apply.

3.2 Implications

FTL motion would allow observers to disagree on event ordering, potentially violating causality. Standard clocks fail, and temporal ordering becomes observer-dependent.

4. Alternative Ways to Think About Time

4.1 Emergent or Relational Time

Time may emerge from correlations between physical systems rather than existing as a fundamental dimension. This view is explored in relational and quantum gravity approaches.

4.2 Entropy or Information Time

Another approach links time to entropy increase or information flow, offering a possible ordering mechanism even when proper time is undefined.

4.3 Tachyons

Hypothetical tachyons are particles that always travel faster than light. While unphysical, they serve as useful conceptual tools for examining causality.

5. Comparative Table

Newtonian Time: Absolute

Relativistic Time: Velocity-dependent

Quantum Time: External parameter

FTL Time (hypothetical): Undefined / imaginary

Emergent Time: Context-dependent

6. Discussion

The consideration of faster-than-light motion as a thought experiment highlights the extent to which contemporary concepts of time are inseparable from the physical constraints imposed by relativity. Within standard relativistic frameworks, time is defined operationally through proper time measured along time like worldlines. When motion is restricted to velocities below the speed of light, this definition remains consistent and observer-independent within well-defined limits. However, hypothetical faster-than-light trajectories fall outside this domain, rendering proper time undefined and undermining the applicability of conventional clock-based descriptions.

This breakdown suggests that time, as employed in current physical theories, may function less as a fundamental entity and more as a descriptive construct that depends on the causal structure of spacetime. When the light-speed limit is treated as an organizing principle, temporal ordering, causality, and simultaneity acquire precise meanings. Once that limit is conceptually exceeded, these meanings become ambiguous or observer-dependent, indicating that standard formulations of time are contingent upon the same constraints that define relativistic spacetime itself.

From this perspective, faster-than-light thought experiments do not merely introduce paradoxes but serve as diagnostic tools for identifying the assumptions embedded in prevailing theories of time. They motivate consideration of alternative frameworks in which temporal ordering might arise from correlations between physical systems, entropy production, or information flow rather than from proper time along worldlines. While such approaches remain speculative, they underscore the possibility that time may emerge from deeper relational or statistical structures rather than existing as an independent, universal parameter.

Ultimately, the value of examining faster-than-light scenarios lies not in their physical plausibility, but in their ability to expose the conceptual boundaries of existing theories. By exploring where and how standard definitions of time fail, such thought experiments contribute to a clearer understanding of what time represents within current physics—and where new conceptual tools may be required.

7. Conclusion

Although faster-than-light travel remains speculative and unsupported by current physical theory, considering its implications provides a useful lens through which to examine the conceptual limits of established frameworks. By treating faster-than-light motion as a thought experiment, this paper has highlighted how central notions such as proper time, causal ordering, and temporal measurement depend critically on the light-speed constraint embedded in relativity.

The breakdown of standard definitions of time in hypothetical faster-than-light scenarios suggests that time, as presently formulated, may not be a universally fundamental quantity

but rather a construct shaped by specific physical conditions. This observation encourages reconsideration of whether time should be regarded as an intrinsic feature of reality or as an emergent descriptor arising from spacetime structure, correlations between systems, or informational and entropic processes.

While no claim is made regarding the physical realizability of faster-than-light motion, the analysis underscores the value of conceptual exploration in theoretical physics. Thought experiments of this kind help delineate the scope of existing theories and clarify the assumptions upon which they rest. In doing so, they open space for alternative perspectives on how events may be ordered and how temporal structure might be understood when conventional limits are conceptually exceeded.

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